The normalized orbit budget at odd degree

Human Compression project: L044 continuation

Retain a transitive connected string C-marking $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le \mathrm{Alt}(\Omega)$, all ambient maximal omissions intransitive, $r \ge 6$ and $n = 2r+1$. Assume the actual transitive six-point head on $D = \{1, \dots, 6\}$ and the literal interface

$$\tau = \rho_3 = (6\ b)\delta, \quad R = G_{\ge 3} = \mathrm{Alt}(\Omega \setminus \{1, \dots, 5\}), \quad \sigma = \rho_4, \quad \sigma|_D = 1.$$

The odd involution δ fixes D, b. Write $H = G_{\ge 5}$ and $K = G_{>5}$, with H fixing D, b. Suppose every tail colour has two ambient-eligible edges in one actual H-orbit. Normalize these assignments by the complete whole-orbit transport. In particular K is intransitive on every orbit with nonempty assigned set.

Theorem 1. *Under these hypotheses* $|\operatorname{Fix}(H)| \ge 9$. *Thus* $x = 4$ *supplies the local inequality* $|\operatorname{Fix}(G_{>x})| \ge 2x + 1$.

Proof. The acquired eight-fixed-point theorem excludes exactly eight fixed points, and the standing interface gives at least seven. It remains to exclude exactly seven. Suppose this occurs. For each nontrivial H-orbit O_i put $d_i = |F_i| - |O_i|/2$, and let p index the orbit whose assigned set contains 5. The acquired normalized deficit theorem gives

$$\sum_i d_i = -2, \quad d_p \le 0, \quad d_i \le -1 \ (|O_i| \text{ even}), \quad d_i \le -\tfrac{1}{2} \ (|O_i| \text{ odd}) \quad (i \ne p). \tag{1}$$

Empty assignments have their exact deficit $-|O_i|/2$. Every nontrivial internal action of δ is free, because it centralizes the transitive group on that orbit. On an orbit of size $2q$ the pair-forest bound therefore gives deficit at most -1 for a nonempty assignment. The same bound holds for an empty assignment; if $q = 1$ the assignment is necessarily empty, because $\tau \in G_j$ already joins its two points for every tail omission.

First δ preserves every nontrivial H-orbit. Indeed, if it exchanges orbits of size m, transport all assigned pairs of one orbit to the other by the actual τ. The complete selected-colour bound gives combined deficit at most $-\lceil m/2 \rceil$. For $m = 2$ or 3 neither orbit contains two edges of an involution, so both assignments are empty; their combined deficit is $-m$. An exchange with odd m therefore costs at least three deficit units and contradicts (1). Every exchange of even orbits costs at least two units and is itself even. Since δ fixes the seven fixed points and is odd, some further odd internal action would be required; it costs at least one more unit. The other deficits are nonpositive, so even exchanges also contradict (1).

It follows that δ acts oddly on some preserved orbit O of size $2q$, with q odd. The actual head argument supplies

$$z = b^\sigma \ne b, \qquad \delta z = z, \qquad Kz = z, \quad z \notin D \cup \{b\}. \tag{2}$$

In particular z cannot belong to O.

We first dispose of every nonfirst saturated internal orbit of size $2q \ge 6$, irrespective of the parity of q. Its deficit is -1 and its assigned set has $q - 1$ colours. The complete odd-degree row theorem gives a terminal assigned label $t < r - 1$ with $|\operatorname{Fix}(G_{>t})| \ge 2t + 3$. The nonempty remaining interval still has ambient eligible pairs for each of its colours. Its complete interval bound gives support at least $2(r - 1 - t) + 1$. Together these force $n \ge 2r + 2$, impossible.

Suppose now $q \ge 3$ for the chosen odd orbit O. An empty assignment would have deficit $-q \le -3$; thus it is nonempty. If $O \ne O_p$, this excludes deficit -1. Equation (1) leaves only deficit

-2, and all other deficits must vanish. Hence there is precisely one further orbit $V = O_p$, of even size $2s$ and deficit zero. Its two assigned 5-edges force $s \geq 2$. This is exactly the acquired nonfirst internal deficit-two theorem, which excludes the configuration.

If $O = O_p$, its integral deficit is -1 or -2. Deficit -2 would leave no further orbit by (1), contradicting (2). With deficit -1, the remaining orbits are either one even orbit V of deficit -1, or two odd orbits V, W each of deficit $-1/2$. The first alternative is exactly the acquired first-internal theorem, including its two-point residual, and is impossible. In the second alternative both remaining assigned sets are nonempty. The complete internal pair-path equality gives an initial interval for F_O, and the complementary terminal interval $I = F_V \cup F_W$ has all its eligible edges confined to $V \cup W$: on O its pair images are trivial, so its possible edges there are already joined by τ in their ambient omissions. Put $T = G_I$. Normalization makes K intransitive on each of V, W, and $T \leq K$, so $o_T(V \cup W) \geq 4$. Complete localization gives

$$|V| + |W| - 2 = 2|I| \leq |V| + |W| - o_T(V \cup W) \leq |V| + |W| - 4,$$

a contradiction. Thus all possibilities with $q \geq 3$ have been excluded.

Finally let $q = 1$, so $O = E$ is an empty assigned pair of deficit -1. If δ fixes every other orbit pointwise, it is literally one transposition, and the acquired exterior-pair theorem applies with all its natural-R, head and ambient-pair hypotheses. It gives a contradiction. Otherwise let W be a further orbit on which δ acts nontrivially. Its deficit is at most -1, hence exactly -1 by (1). If $|W| = 2$, its assignment is also empty. All remaining deficit is zero, so the only further orbit is O_p, where δ must be trivial. The two transpositions on E, W would make δ even, a contradiction.

We may therefore suppose $|W| = 2q' \geq 4$. Its assigned set is nonempty, because an empty one costs at least two units. If $W = O_p$, (1) leaves no other orbit besides E, W. Both are free under δ, again contradicting (2). If $W \neq O_p$, every remaining deficit is zero and there is precisely one further orbit $V = O_p$, of even size and deficit zero. For $q' \geq 3$, the saturated nonfirst exclusion above applies. For $q' = 2$, the literal budget is

$$|E| = 2, \quad |W| = 4, \quad F_E = \varnothing, \quad |F_W| = 1, \quad F_V = \{5, \ldots, r-1\} \setminus F_W.$$

The involution δ is free on E, W and fixes V pointwise by (2) and centralization. This is the full four-point budget excluded by the acquired fixed-anchor corollary. It is impossible as well. Every case with seven fixed points has now been excluded. Together with the acquired eight-point exclusion this proves the theorem. □

Attribution and boundary. The deficit, transport, internal pair-path, row-growth, localization, fixed-anchor and residual orbit proofs are acquired in full. This assembly supplies the exhaustive seven-fixed-point case division and states its local consequence. It does not produce an eligible pair in one orbit when the two edges are split between different orbits, and it does not assert that later ambient generators preserve every locally fixed point. Those assignment and attachment obligations remain necessary for complete L044. No new rank induction, computation, formal, original or question credit.